MD. ROBIUL HASSAN

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from BRAC University (CGPA 3.50/4.00, eligible for Distinction Award upon degree completion) with proven expertise in full-stack web engineering and applied machine learning research. Successfully defended undergraduate thesis developing a multimodal mental health assessment framework for rural Bangladeshi women, achieving 94.12% accuracy via multilingual text embeddings and SVM. Demonstrates industry experience through a web development internship at GAOTek Inc. and multiple production-grade full-stack software projects, building high-performance web applications, RESTful APIs, and database systems. Seeking admission to a Master of Computer Science program at top-ranked universities to advance research in intelligent systems, machine learning, and scalable software architectures.

EDUCATION

Bachelor of Science in Computer Science and Engineering — May 2022 – August 2026
BRAC University, Dhaka, Bangladesh — CGPA: **3.50 / 4.00**

• Academic Honor: Distinction Award — To be awarded upon degree completion (CGPA 3.50 – 3.64 tier)

Research Interests: Applied Machine Learning, Natural Language Processing, Speech & Acoustic Analysis, Multimodal Systems, Distributed Web Engineering

Higher Secondary Certificate (HSC) — Science — 2019 – 2021
Udayan Uchcha Madhyamik Bidyalaya, Dhaka (Dhaka Board) — GPA: **4.83 / 5.00**

Secondary School Certificate (SSC) — Science — 2017 – 2019
Bright High School, Dhaka (Dhaka Board) — GPA: **5.00 / 5.00**

IELTS Academic — Overall Band: 7.0 — October 2026
Listening: 8.5 • **Reading:** 6.5 • **Writing:** 6.0 • **Speaking:** 6.0

UNDERGRADUATE THESIS & RESEARCH

A Multimodal System to Support the Mental Health of Rural Woman — May 2025 – August 2026
*Department of Computer Science and Engineering, BRAC University (Status: **Defended**)*

Supervisors: Dr. Amitabha Chakrabarty (Professor) & Dewan Ziaul Karim (Senior Lecturer)

- Formulated a machine-learning framework designed to support depression and anxiety risk assessment among rural Bangladeshi women utilizing both vocal speech and transcribed text features.
- **Best Depression Model:** Achieved **94.12% accuracy**, **93.77% macro-F1**, and **1.00 ROC-AUC** using **Multilingual E5 text embeddings with Linear SVM**.
- **Multimodal Anxiety Fusion:** Developed late-fusion architecture integrating text and speech probability predictions, attaining **91.18% accuracy**, **94.00% balanced accuracy**, and **89.67% macro-F1**.
- **Acoustic Feature Extraction:** Extracted clinical vocal features using the **eGeMAPS** acoustic standard, achieving **88.24% accuracy and 86.56% macro-F1** with RBF SVM on voice-only recordings.
- **Pipeline Engineering:** Evaluated diverse classification benchmarks (Linear/RBF SVM, Random Forest, Gradient Boosting, KNN, Logistic Regression) and resolved clinical data class imbalances via **SMOTE**.

TECHNICAL SKILLS

Programming Languages:	Python, JavaScript (ES6+), TypeScript, C/C++, PHP, SQL
Machine Learning & NLP:	Scikit-learn, Multilingual E5 Embeddings, eGeMAPS Acoustic Extraction, TF-IDF, SMOTE, Classification (SVM, RF, Gradient Boosting, KNN, Logistic Regression), Feature Engineering
Web & Full-Stack:	React.js, Next.js, React Native, Node.js, Express.js, RESTful APIs, oRPC, HTML5, CSS3, Tailwind CSS, Vite
Databases & Tools:	PostgreSQL, MongoDB, Mongoose, MySQL, Git, GitHub, Docker, Postman, VS Code, Linux
Core CS Concepts:	Data Structures & Algorithms, Object-Oriented Design (OOP), Software Architecture, Operating Systems, Computer Graphics, Computer Networks

PROFESSIONAL EXPERIENCE

GAOTek Inc. January 22, 2024 – April 22, 2024

Web Development Intern Remote (Ontario, Canada)

- Optimized and maintained enterprise web pages on **WordPress CMS**, improving page load performance and responsive rendering across desktop, tablet, and mobile viewports.
- Developed reusable frontend UI components and responsive page layouts utilizing **HTML5, CSS3, JavaScript, and Bootstrap**, adhering to web accessibility (WCAG) standards.
- Applied **SEO best practices** and image/asset optimization techniques, contributing to measurable improvements in search engine visibility and Core Web Vitals scores.
- Collaborated with a cross-functional international team using **Git** version control and agile workflows, participating in code reviews, sprint planning, and iterative deployment cycles.

SELECTED SOFTWARE & ACADEMIC PROJECTS

Bright Tutor — Tuition Booking Platform [Live Site | GitHub] 2026

Tech Stack: Next.js, React, TypeScript, oRPC, Tailwind CSS, PLpgSQL, Docker

- Built a comprehensive tuition management ecosystem connecting guardians, students, tutors, and administrators with role-based onboarding and Google authentication.
- Implemented type-safe end-to-end communication using oRPC, robust relational schemas in PLpgSQL, and asynchronous messaging architecture.

RestoFlow — Restaurant Management & POS [Live Site | Video Demo | GitHub] CSE471 Architecture Project

Tech Stack: JavaScript, React, Node.js, Express.js, WebSockets, MongoDB, Cloudinary

- Designed full-stack dining platform featuring QR-code table detection, bilingual menu (English/Bangla), dietary taste filtering, and live kitchen order broadcasts via WebSockets.
- Engineered Role-Based Access Control (RBAC) for admins, waiters, and kitchen staff alongside an integrated POS engine supporting recursive bill splitting and discount codes.

SelfShop — Commercial E-Commerce Platform [Live Site | GitHub] 2025 – 2026

Tech Stack: PHP, MySQL, JavaScript, HTML5, CSS3, REST APIs

- Engineered production commercial e-commerce platform supporting product cataloging, category management, dynamic cart operations, and responsive checkout workflows.
- Developed administrative vendor and inventory management modules with automated order tracking and secure SQL data operations.

Snowboard Rush — 3D Graphics Downhill Game CSE423 Computer Graphics Project

Tech Stack: Python, PyOpenGL, 3D Mathematics

- Engineered a 3D downhill endless-runner in Python with PyOpenGL; implemented custom perspective camera projection, collision detection, and continuous downhill velocity physics.
- Developed procedural mountain hazards (static trees, breakable trees, boulders, flying birds), atmospheric foggy weather shaders, aerial jump/spin acrobatics, and multi-life gameplay logic.

ACADEMIC HIGHLIGHTS & CORE COURSEWORK

Core CS Coursework:	Data Structures, Algorithms, Software Architecture (CSE471), Computer Graphics (CSE423), Operating Systems, Database Management Systems, Artificial Intelligence, Computer Networks
Additional Work:	Wholesale B2B Platform (GitHub), Bikalpo Wholesale Grocery (bikalpo.com | GitHub), DineEase Restaurant Reservation (GitHub), 3D Enemy Shooter (PyOpenGL), DIMS E-Commerce (rangpurelectrohub.com | GitHub), GymFit Web Portal (GitHub), Rockdale Real Estate (GitHub), Tic-Tac-Toe (GitHub)

REFERENCE

Dr. Amitabha Chakrabarty Undergraduate Thesis Supervisor & Recommender

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